

Challenge (Habanero)

- Q1.** A gamer's score, x , satisfies $|x| = 25$. What are the possible values of x ?
- (A) $x = 25$
(B) $x = -25$
(C) $x = 25$ or $x = -25$
(D) $x = 0$
(E) $x = 12.5$
- Q2.** Solve $|2x - 3| = 7$ for x .
- (A) $x = 2$ or $x = 5$
(B) $x = -2$ or $x = -5$
(C) $x = 2$ or $x = -5$
(D) $x = 5$ or $x = -2$
(E) $x = 1$ or $x = -1$
- Q3.** What is the solution to $|x + 2| = 9$?
- (A) $x = 7$
(B) $x = -7$
(C) $x = 7$ or $x = -7$
(D) $x = 11$
(E) $x = -11$
- Q4.** A coding error causes a program to malfunction when $|3x - 1| \leq 5$. What values of x will cause the malfunction?
- (A) $-2 \leq x \leq 2$
(B) $-1 \leq x \leq 2$
(C) $-1.67 \leq x \leq 2$
(D) $0 \leq x \leq 2$
(E) $-4 \leq x \leq 2$
- Q5.** Solve $|x| - 2 = 3$ for x .
- (A) $x = 5$
(B) $x = -5$
(C) $x = 5$ or $x = -5$
(D) $x = 3$
(E) $x = -3$
- Q6.** What values of x satisfy $|2x + 1| = |x - 3|$?
- (A) $x = 1$
(B) $x = -1$
(C) $x = 1$ or $x = -4$
(D) $x = -1$ or $x = 4$
(E) $x = 2$
- Q7.** A gamer's high score, x , is given by $|x - 10| = 5$. What are the possible values of x ?
- (A) $x = 5$
(B) $x = 15$
(C) $x = 5$ or $x = 15$
(D) $x = 0$
(E) $x = 20$
- Q8.** Solve $|x^2 - 4| = 3$ for x .
- (A) $x = 1$ or $x = -1$
(B) $x = \sqrt[pm]{7}$
(C) $x = \sqrt[pm]{1}$
(D) $x = 2$ or $x = -2$
(E) $x = \sqrt[pm]{1}$

Open Ended Questions (FRQ)

- Q9.** Solve $|x + 1| = |2x - 3|$ and check for extraneous solutions.
- Q10.** A programming team needs to solve $|3x - 2| = |x + 4|$ to debug their code. Find the values of x that satisfy the equation and explain your steps.

Chef's Tips

- Tip 1: Encourage students to check their solutions for extraneous results.
- Tip 2: Remind students to consider multiple scenarios when solving absolute value equations.
- Tip 3: Suggest using graphical methods to visualize the solutions to absolute value equations.